

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5281

5 Credits: 4

Program Core

Source-grounded

Object Oriented Programming

- Understand object-oriented Principles.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5281 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5281.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Computer Engineering / Computer Hardware Engineering/
Course code	5281
Course title	Object Oriented Programming
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

31 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- ● Understand object-oriented Principles.
- ● Write, Compile and Execute Java Programs.
- ● Familiarize GUI programming using Swing.
- ● Introduce database connectivity for developing applications.

Course outcomes

CO	Official outcome	Study level
CO1	16 Applying develop simple JAVA applications. Build classes using inheritance, interfaces and	See official syllabus
CO2	16 Applying packages	See official syllabus
CO3	Build GUI applications using swing. 14 Applying	See official syllabus
CO4	Develop applications using Database. 12 Applying	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	applications	12	As specified
Module 2	Build classes using inheritance, interfaces and packages.	8	As specified
Module 3	Build GUI applications using swing.	6	As specified
Module 4	Develop applications using database	5	As specified

MODULE 1

applications

This module is presented from the official syllabus scope for Object Oriented Programming. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: applications
- Official duration: As specified
- Official topic entries captured: 12
- The full source excerpt is preserved below for coverage checking.

▼ 1 Understanding. Programming. Explain the JDK & IDE's for developing java

1 Understanding. Programming. Explain the JDK & IDE's for developing java is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding. Programming. Explain the JDK & IDE's for developing java using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding. programming. explain the jdk & ide's for developing java should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 1 Understanding. Programming. Explain the JDK & IDE's for developing java means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 1 Understanding. Programming. Explain the JDK & IDE's for developing java definition/meaning. steps, application. Technical terms English- .

▼ 1 Understanding. applications Outline the general structure of a Java Understanding.

1 Understanding. applications Outline the general structure of a Java Understanding. is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding. applications Outline the general structure of a Java Understanding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding. applications outline the general structure of a java understanding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-എന്ന 1 Understanding. applications Outline
the general structure of a Java Understanding. ഏകദേശം എത്രയെന്ന്
definition/meaning എത്രയെന്ന്. എത്രയെന്ന് എത്രയെന്ന്, steps,
application എത്രയെന്ന് എത്രയെന്ന്. Technical terms English-ൽ എത്രയെന്ന്
എത്രയെന്ന്.

▼ program and explain steps to create, compile 1 and execute a program Describe objects and classes and use classes

program and explain steps to create, compile 1 and execute a program Describe objects and classes and use classes is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify program and explain steps to create, compile 1 and execute a program Describe objects and classes and use classes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, program and explain steps to create, compile 1 and execute a program describe objects and classes and use classes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What program and explain steps to create, compile 1 and execute a program Describe objects and classes and use classes means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പ്രോഗ്രാം ചെയ്യാനും പ്രവർത്തിക്കാനും എങ്ങനെ ചെയ്യണം എന്ന് വിശദീകരിക്കുക. പ്രോഗ്രാം ചെയ്യാനും പ്രവർത്തിക്കാനും എങ്ങനെ ചെയ്യണം എന്ന് വിശദീകരിക്കുക. definition/meaning വിശദീകരിക്കുക. പ്രോഗ്രാം ചെയ്യാനും പ്രവർത്തിക്കാനും എങ്ങനെ ചെയ്യണം എന്ന് വിശദീകരിക്കുക. steps, application വിശദീകരിക്കുക. Technical terms English-ൽ വിശദീകരിക്കുക.

▼ 2 Understanding. to model objects. Demonstrate how to define classes and create

2 Understanding. to model objects. Demonstrate how to define classes and create is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding. to model objects. Demonstrate how to define classes and create using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding. to model objects. demonstrate how to define classes and create should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 2 Understanding. to model objects. Demonstrate how to define classes and create means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Understanding. to model objects. Demonstrate how to define classes and create definition/meaning. , steps, application . Technical terms English- .

▼ 1 Applying. objects

1 Applying. objects is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Applying. objects using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 applying. objects should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Applying. objects means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പാഠ്യ 1 Applying. objects പദപരിചയം/അർത്ഥം definition/meaning പദപരിചയം/അർത്ഥം. പദപരിചയം പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English-ൽ പദപരിചയം പദപരിചയം/അർത്ഥം.

▼ Create Objects using constructors. 1 Applying Interpret accessing objects via object

Create Objects using constructors. 1 Applying Interpret accessing objects via object is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Create Objects using constructors. 1 Applying Interpret accessing objects via object using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, create objects using constructors. 1 applying interpret accessing objects via object should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Create Objects using constructors. 1 Applying Interpret accessing objects via object means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Create Objects using constructors. 1 Applying Interpret accessing objects via object definition/meaning, steps, application Technical terms English-

▼ 1 Understanding. reference variables. Distinguish between instance and static

1 Understanding. reference variables. Distinguish between instance and static is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding. reference variables. Distinguish between instance and static using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding. reference variables. distinguish between instance and static should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 1 Understanding. reference variables. Distinguish between instance and static means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 1 Understanding. reference variables. Distinguish between instance and static definition/meaning. , steps, application . Technical terms English- .

▼ 1 Understanding. variables and methods. Develop Java Programs to implement classes

1 Understanding. variables and methods. Develop Java Programs to implement classes is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding. variables and methods. Develop Java Programs to implement classes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding. variables and methods. develop java programs to implement classes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 1 Understanding. variables and methods. Develop Java Programs to implement classes means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 1 Understanding. variables and methods. Develop Java Programs to implement classes definition/meaning . . . , steps, application Technical terms English-

▼ 2 Applying. and objects. Explain Visibility Modifiers and Data

2 Applying. and objects. Explain Visibility Modifiers and Data is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying. and objects. Explain Visibility Modifiers and Data using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying. and objects. explain visibility modifiers and data should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying. and objects. Explain Visibility Modifiers and Data means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-2 Applying. and objects. Explain Visibility Modifiers and Data definition/meaning.
 definition/meaning, steps, application
 Technical terms English-

▼ 1 Understanding. Encapsulation Develop methods with object arguments an

1 Understanding. Encapsulation Develop methods with object arguments an is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding. Encapsulation Develop methods with object arguments an using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding. encapsulation develop methods with object arguments an should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 1 Understanding. Encapsulation Develop methods with object arguments an means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 1 Understanding. Encapsulation Develop methods with object arguments an definition/meaning . . . , steps, application Technical terms English-

▼ differentiate between Primitive arguments 2 Applying and object type arguments

differentiate between Primitive arguments 2 Applying and object type arguments is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify differentiate between Primitive arguments 2 Applying and object type arguments using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, differentiate between primitive arguments 2 applying and object type arguments should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What differentiate between Primitive arguments 2 Applying and object type arguments means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- $\frac{1}{2}$ differentiate between Primitive arguments 2 Applying and object type arguments $\frac{1}{2}$ definition/meaning $\frac{1}{2}$ application $\frac{1}{2}$ steps, application $\frac{1}{2}$ Technical terms English- $\frac{1}{2}$ $\frac{1}{2}$.

▼ **Explain Exception Handling. 2 Understanding Contnets: Object Oriented Programming (OOP) - Characteristics of OOP - Features of JAVA - Advantages of JAVA - Tools Available of JAVA Programming (JDK, JAVA Packages, various IDEs like NetBeans, eclipse) - Building Java applications. Objects and Classes - Defining a Class - Declaring attributes, Declaring and Defining methods, Creating Object - Accessing Objects - Constructors - Constructor overloading - Static variables, constants and methods - method overloading - Visibility modifiers, Data field Encapsulation, passing and returning objects as arguments, Array of objects-Exception handling, Try,catch,-Multiple catch & finally statement.**

Explain Exception Handling. 2 Understanding Contnets: Object Oriented Programming (OOP) - Characteristics of OOP - Features of JAVA - Advantages of JAVA - Tools Available of JAVA Programming (JDK, JAVA Packages, various IDEs like NetBeans, eclipse) - Building Java applications. Objects and Classes - Defining a Class - Declaring attributes, Declaring and Defining methods, Creating Object - Accessing Objects - Constructors - Constructor overloading - Static variables, constants and methods - method overloading - Visibility modifiers, Data field Encapsulation, passing and returning objects as arguments, Array of objects-Exception handling, Try,catch,-Multiple catch & finally statement. is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Exception Handling. 2 Understanding Contnets: Object Oriented Programming (OOP) – Characteristics of OOP – Features of JAVA – Advantages of JAVA – Tools Available of JAVA Programming (JDK, JAVA Packages, various IDEs like NetBeans, eclipse) – Building Java applications. Objects and Classes – Defining a Class – Declaring attributes, Declaring and Defining methods, Creating Object – Accessing Objects - Constructors – Constructor overloading – Static variables, constants and methods – method overloading – Visibility modifiers, Data field Encapsulation, passing and returning objects as arguments, Array of objects-Exception handling, Try,catch,-Multiple catch & finally statement. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain exception handling. 2 understanding contnets: object oriented programming (oop) – characteristics of oop – features of java – advantages of java – tools available of java programming (jdk, java packages, various ides like netbeans, eclipse) – building java applications. objects and classes – defining a class – declaring attributes, declaring and defining methods, creating object – accessing objects - constructors – constructor overloading – static variables, constants and methods – method overloading – visibility modifiers, data field encapsulation, passing and returning objects as arguments, array of objects-exception handling, try,catch,-multiple catch & finally statement. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Exception Handling. 2 Understanding Contnets: Object Oriented Programming (OOP) – Characteristics of OOP – Features of JAVA – Advantages of JAVA – Tools Available of JAVA Programming (JDK, JAVA

Aspect	What to prepare
	Packages, various IDEs like NetBeans, eclipse) – Building Java applications. Objects and Classes – Defining a Class – Declaring attributes, Declaring and Defining methods, Creating Object – Accessing Objects - Constructors – Constructor overloading – Static variables, constants and methods – method overloading – Visibility modifiers, Data field Encapsulation, passing and returning objects as arguments, Array of objects-Exception handling, Try,catch,-Multiple catch & finally statement. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain Exception Handling. 2 Understanding Contnets: Object Oriented Programming (OOP) – Characteristics of OOP – Features of JAVA – Advantages of JAVA – Tools Available of JAVA Programming (JDK, JAVA Packages, various IDEs like NetBeans, eclipse) – Building Java applications. Objects and Classes – Defining a Class – Declaring attributes, Declaring and Defining methods, Creating Object – Accessing Objects - Constructors – Constructor overloading – Static variables, constants and methods – method overloading – Visibility modifiers, Data field Encapsulation, passing and returning objects as arguments, Array of objects-Exception handling, Try,catch,-Multiple catch & finally statement. definition/meaning. steps, application. Technical terms English-

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Build classes using inheritance, interfaces and packages.

This module is presented from the official syllabus scope for Object Oriented Programming. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Build classes using inheritance, interfaces and packages.
- Official duration: As specified
- Official topic entries captured: 8
- The full source excerpt is preserved below for coverage checking.

▼ 3 Understanding inheritance with examples. Illustrate invoking the superclass constructor

3 Understanding inheritance with examples. Illustrate invoking the superclass constructor is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding inheritance with examples. Illustrate invoking the superclass constructor using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding inheritance with examples. illustrate invoking the superclass constructor should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Understanding inheritance with examples. Illustrate invoking the superclass constructor means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Understanding inheritance with examples. Illustrate invoking the superclass constructor
 definition/meaning .
 , steps, application . Technical terms English-
 .

▼ 2 Applying and methods using super keywords. Illustrate overriding instance methods in the

2 Applying and methods using super keywords. Illustrate overriding instance methods in the is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying and methods using super keywords. Illustrate overriding instance methods in the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying and methods using super keywords. illustrate overriding instance methods in the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 2 Applying and methods using super keywords. Illustrate overriding instance methods in the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 2 Applying and methods using super keywords. Illustrate overriding instance methods in the definition/meaning. 2 Applying and methods using super keywords, steps, application. Technical terms English- 2 Applying and methods using super keywords.

▼ 2 Applying subclass.

2 Applying subclass. is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying subclass. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying subclass. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying subclass. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 2 Applying subclass.
 definition/meaning . steps, application . Technical terms English-
 .

▼ Illustrate polymorphism and dynamic binding

Illustrate polymorphism and dynamic binding is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate polymorphism and dynamic binding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate polymorphism and dynamic binding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate polymorphism and dynamic binding means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഇംഗ്ലീഷ് ഉദാഹരണം പോലുള്ളവയെക്കുറിച്ച് വിശദീകരിക്കുക. ഇംഗ്ലീഷ് വിശദീകരണം, steps, application എന്നിവ ഉൾപ്പെടെയുള്ളവയെക്കുറിച്ച് വിശദീകരിക്കുക. Technical terms English-ൽ ഉൾപ്പെടെയുള്ളവയെക്കുറിച്ച് വിശദീകരിക്കുക.

▼ Explain visibility controls 1 Understanding Build abstract class, final class and method

Explain visibility controls 1 Understanding Build abstract class, final class and method is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain visibility controls 1 Understanding Build abstract class, final class and method using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain visibility controls 1 understanding build abstract class, final class and method should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What Explain visibility controls 1 Understanding Build abstract class, final class and method means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Explain visibility controls 1
Understanding Build abstract class, final class and method
definition/meaning . steps, application . Technical terms English-
.

▼ 2 Applying overriding

2 Applying overriding is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying overriding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying overriding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying overriding means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പാഠ്യ 2 Applying overriding പദപരിവർത്തനം പദപരിവർത്തനം definition/meaning പദപരിവർത്തനം. പദപരിവർത്തനം പദപരിവർത്തനം, steps, application പദപരിവർത്തനം പദപരിവർത്തനം പദപരിവർത്തനം. Technical terms English-ൽ പദപരിവർത്തനം പദപരിവർത്തനം.

▼ Develop interfaces

Develop interfaces is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop interfaces using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop interfaces should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop interfaces means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എം വികസിപ്പിക്കുക ഇന്റർഫേസ് നിർവ്വചനം/അർത്ഥം നിർവ്വചിക്കുക. നിർവ്വചനം നിർവ്വചിക്കുക, steps, application നിർവ്വചിക്കുക നിർവ്വചിക്കുക. Technical terms English-ൽ നിർവ്വചിക്കുക നിർവ്വചിക്കുക.

▼ Develop packages 2 Applying Inheritance - Types of Inheritance - Superclass and subclass - calling super class constructor and methods - overloading vs overriding - Final variables, final methods and final classes - Abstract methods and class - polymorphism - dynamic binding - Visibility controls. Interfaces - Definition - Extending, Implementing and accessing Interfaces- packages - Creating and Accessing of Packages.

Develop packages 2 Applying Inheritance - Types of Inheritance - Superclass and subclass - calling super class constructor and methods - overloading vs overriding - Final variables, final methods and final classes - Abstract methods and class - polymorphism - dynamic binding - Visibility controls. Interfaces - Definition - Extending, Implementing and accessing Interfaces- packages - Creating and Accessing of Packages. is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop packages 2 Applying Inheritance - Types of Inheritance - Superclass and subclass - calling super class constructor and methods - overloading vs overriding - Final variables, final methods and final classes - Abstract methods and class - polymorphism - dynamic binding - Visibility controls. Interfaces - Definition - Extending, Implementing and accessing Interfaces- packages - Creating and Accessing of Packages. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop packages 2 applying inheritance – types of inheritance – superclass and subclass – calling super class constructor and methods – overloading vs overriding – final variables, final methods and final classes – abstract methods and class – polymorphism – dynamic binding – visibility controls. interfaces – definition – extending, implementing and accessing interfaces- packages – creating and accessing of packages. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop packages 2 Applying Inheritance – Types of Inheritance – Superclass and subclass – calling super class constructor and methods – overloading vs overriding – Final variables, final methods and final classes – Abstract methods and class – polymorphism – dynamic binding – Visibility controls. Interfaces – Definition – Extending, Implementing and accessing Interfaces- packages – Creating and Accessing of Packages. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Develop packages 2 Applying Inheritance - Types of Inheritance - Superclass and subclass - calling super class constructor and methods - overloading vs overriding - Final variables, final methods and final classes - Abstract methods and class - polymorphism - dynamic binding - Visibility controls. Interfaces - Definition - Extending, Implementing and accessing Interfaces- packages - Creating and Accessing of Packages. പരമ്പരപ്പെടുത്തലിന്റെ നിർവ്വചന/അർത്ഥം പരമ്പരപ്പെടുത്തലിന്റെ തരങ്ങൾ, steps, application പരമ്പരപ്പെടുത്തലിന്റെ പ്രയോഗം. Technical terms English-ൽ പരമ്പരപ്പെടുത്തലിന്റെ.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Build GUI applications using swing.

This module is presented from the official syllabus scope for Object Oriented Programming. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Build GUI applications using swing.
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

▼ Outline GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI

Outline GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI is an official topic in Module 3 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline gui programming in java using swing. 2 understanding. make use of packages containing gui should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Outline GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI
 പാക്കേജുകൾ ഉപയോഗിച്ച് GUI പ്രോഗ്രാമിംഗ്. definition/meaning. പാക്കേജുകൾ, steps, application. Technical terms English-ൽ പാക്കേജുകൾ ഉപയോഗിച്ച്.

▼ containers, components, event-handling classes 2 Applying. and interfaces to develop simple applications.

containers, components, event-handling classes 2 Applying. and interfaces to develop simple applications. is an official topic in Module 3 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify containers, components, event-handling classes 2 Applying. and interfaces to develop simple applications. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, containers, components, event-handling classes 2 applying. and interfaces to develop simple applications. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What containers, components, event-handling classes 2 Applying. and interfaces to develop simple applications. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ containers, components, event-handling classes 2 Applying. and interfaces to develop simple applications.

000000000000 00000 definition/meaning 000000000000. 000000 000000
 00000000, steps, application 000000 0000000000 000000. Technical terms
 English-0 00000 000000000000.

▼ Describe steps to create and manipulate GUI 3 Understanding. controls. Explain handling of mouse events and

Describe steps to create and manipulate GUI 3 Understanding. controls. Explain handling of mouse events and is an official topic in Module 3 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe steps to create and manipulate GUI 3 Understanding. controls. Explain handling of mouse events and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe steps to create and manipulate gui 3 understanding. controls. explain handling of mouse events and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe steps to create and manipulate GUI 3 Understanding. controls. Explain handling of mouse events and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Describe steps to create and manipulate GUI 3 Understanding. controls. Explain handling of mouse events and definition/meaning. , steps, application . Technical terms English- .

▼ 2 Understanding. keyboard events. Explain Common GUI event type and Listener

2 Understanding. keyboard events. Explain Common GUI event type and Listener is an official topic in Module 3 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding. keyboard events. Explain Common GUI event type and Listener using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding. keyboard events. explain common gui event type and listener should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 2 Understanding. keyboard events. Explain Common GUI event type and Listener means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Understanding. keyboard events. Explain Common GUI event type and Listener definition/meaning. steps, application. Technical terms English-

▼ 2 Understanding. types.

2 Understanding. types. is an official topic in Module 3 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding. types. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding. types. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding. types. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Understanding. types. definition/meaning . steps, application . Technical terms English- .

▼ Develop programs using swing Components 3 Applying. Outline GUI programming - Swing - Swing Components - Simple GUI based input/output programs - GUI application development using Swing components such as JLabel, JTextField, JButton, JList, JCheckBox, JComboBox, JPanel, JTextArea - Event Handling - Common GUI event Types and Containers.

Develop programs using swing Components 3 Applying. Outline GUI programming - Swing - Swing Components - Simple GUI based input/output programs - GUI application development using Swing components such as JLabel, JTextField, JButton, JList, JCheckBox, JComboBox, JPanel, JTextArea - Event Handling - Common GUI event Types and Containers. is an official topic in Module 3 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop programs using swing Components 3 Applying. Outline GUI programming - Swing - Swing Components - Simple GUI based input/output programs - GUI application development using Swing components such as JLabel, JTextField, JButton, JList, JCheckBox, JComboBox, JPanel, JTextArea - Event Handling - Common GUI event Types and Containers. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop programs using swing components 3 applying. outline gui programming – swing – swing components – simple gui based input/output programs - gui application development using swing components such as JLabel, JTextField, JButton, JList, JCheckBox, JComboBox, JPanel, JTextArea – event handling – common gui event types and containers. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop programs using swing Components 3 Applying. Outline GUI programming – Swing – Swing Components – Simple GUI based input/output programs - GUI application development using Swing components such as JLabel, JTextField, JButton, JList, JCheckBox, JComboBox, JPanel, JTextArea – Event Handling – Common GUI event Types and Containers. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Develop programs using swing Components 3 Applying. Outline GUI programming – Swing – Swing Components – Simple GUI based input/output programs - GUI application

development using Swing components such as JLabel, JTextField, JButton, JList, JCheckBox, JComboBox, JPanel, JTextArea – Event Handling – Common GUI event Types and Containers. **പരമ്പരാഗതമായി പരമ്പരാഗതമായി definition/meaning പരമ്പരാഗതമായി. പരമ്പരാഗതമായി പരമ്പരാഗതമായി, steps, application പരമ്പരാഗതമായി പരമ്പരാഗതമായി പരമ്പരാഗതമായി. Technical terms English-പരമ്പരാഗതമായി പരമ്പരാഗതമായി.**

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Develop applications using database

This module is presented from the official syllabus scope for Object Oriented Programming. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Develop applications using database
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

▼ Explain relational database & SQL

Explain relational database & SQL is an official topic in Module 4 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain relational database & SQL using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain relational database & sql should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain relational database & SQL means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain relational database & SQL
 പദപരിചയം definition/meaning. പദപരിചയം, steps, application പദപരിചയം പദപരിചയം. Technical terms
 English- പദപരിചയം പദപരിചയം.

▼ Explain SQL commands.

Explain SQL commands. is an official topic in Module 4 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain SQL commands. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain sql commands. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain SQL commands. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്നു വിശദീകരിക്കുക. SQL കമാന്റുകൾ. ഏതൊരു കമാന്റും വിശദീകരിക്കുക. definition/meaning എഴുതുക. ഏതൊരു കമാന്റും വിശദീകരിക്കുക, steps, application എഴുതുക. Technical terms English-ൽ എഴുതുക. എഴുതുക.

▼ Illustrate the stages in JDBC 2 Understanding Illustrate the interfaces in JDBC -register the

Illustrate the stages in JDBC 2 Understanding Illustrate the interfaces in JDBC -register the is an official topic in Module 4 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the stages in JDBC 2 Understanding Illustrate the interfaces in JDBC -register the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the stages in jdbc 2 understanding illustrate the interfaces in jdbc -register the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What Illustrate the stages in JDBC 2 Understanding Illustrate the interfaces in JDBC -register the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Illustrate the stages in JDBC 2 Understanding Illustrate the interfaces in JDBC -register the definition/meaning . . . , steps, application Technical terms English-

▼ driver- connection- Preparedstatement- 2 Applying Executing statements- Resultset

driver- connection- Preparedstatement- 2 Applying Executing statements- Resultset is an official topic in Module 4 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify driver- connection- Preparedstatement- 2 Applying Executing statements- Resultset using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, driver- connection- preparedstatement- 2 applying executing statements- resultset should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What driver- connection- Preparedstatement- 2 Applying Executing statements- Resultset means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- driver- connection- Preparedstatement- 2 Applying Executing statements- Resultset definition/meaning . . . , steps, application Technical terms English-

▼ **Develop simple application with database 4 Applying Database - Creation of databases-Tables -SQL commands like -create, alter, drop, select, insert, delete Database Connectivity - JDBC - RowSet interface - PreparedStatement - retrieving results - closing connection. Text / Reference T/R Book Title/Author Herbert Schildt, Dale Skrien: Java Fundamentals A Comprehensive T1 Introduction T2 Balaguruswamy E: Programming with Java, 6th edition. R1 Liang, Y Daniel: Introduction to JAVA Programming, Pearson, 9th Ed. R2 Herbert Schildt: Java The Complete Reference, Seventh Edition, R3 Herbert Schildt: Swing A Beginner's guide R4 Paul Deital, Harvey Deital: Java How to Program, R5 Yang HU: Easy Learning JDBC+Mysql Online Resources Sl.No Website Link 1 https://onlinecourses.nptel.ac.in/noc20_cs08/course 2 <https://www.tutorialspoint.com/java>**

Develop simple application with database 4 Applying Database - Creation of databases-Tables -SQL commands like -create, alter, drop, select, insert, delete Database Connectivity - JDBC - RowSet interface - PreparedStatement - retrieving results - closing connection. Text / Reference T/R Book Title/Author Herbert Schildt, Dale Skrien: Java Fundamentals A Comprehensive T1 Introduction T2 Balaguruswamy E: Programming with Java, 6th edition. R1 Liang, Y Daniel: Introduction to JAVA Programming, Pearson, 9th Ed. R2 Herbert Schildt: Java The Complete Reference, Seventh Edition, R3 Herbert Schildt: Swing A Beginner's guide R4 Paul Deital, Harvey Deital: Java How to Program, R5 Yang HU: Easy Learning JDBC+Mysql Online Resources Sl.No Website Link 1 https://onlinecourses.nptel.ac.in/noc20_cs08/course 2 <https://www.tutorialspoint.com/java> is an official topic in Module 4 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or

stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop simple application with database 4 Applying Database – Creation of databases-Tables -SQL commands like -create, alter, drop, select, insert, delete Database Connectivity – JDBC – RowSet interface – PreparedStatement - retrieving results - closing connection. Text / Reference T/R Book Title/Author Herbert Schildt, Dale Skrien: Java Fundamentals A Comprehensive T1 Introduction T2 Balaguruswamy E: Programming with Java, 6th edition. R1 Liang, Y Daniel: Introduction to JAVA Programming, Pearson, 9th Ed. R2 Herbert Schildt: Java The Complete Reference, Seventh Edition, R3 Herbert Schildt: Swing A Beginner's guide R4 Paul Deital, Harvey Deital: Java How to Program, R5 Yang HU: Easy Learning JDBC+Mysql Online Resources Sl.No Website Link 1 https://onlinecourses.nptel.ac.in/noc20_cs08/course 2 <https://www.tutorialspoint.com/java> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop simple application with database 4 applying database – creation of databases-tables -sql commands like -create, alter, drop, select, insert, delete database connectivity – jdbc – rowset interface – preparedstatement - retrieving results - closing connection. text / reference t/r book title/author herbert schildt, dale skrien: java fundamentals a comprehensive t1 introduction t2 balaguruswamy e: programming with java, 6th edition. r1 liang, y daniel: introduction to java programming, pearson, 9th ed. r2 herbert schildt: java the complete reference, seventh edition, r3 herbert schildt: swing a beginner's guide r4 paul deital, harvey deital: java how to program, r5 yang hu: easy learning jdbc+mysql online resources sl.no website link 1

https://onlinecourses.nptel.ac.in/noc20_cs08/course 2

<https://www.tutorialspoint.com/java> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop simple application with database 4 Applying Database – Creation of databases-Tables -SQL commands like -create, alter, drop, select, insert, delete Database Connectivity – JDBC – RowSet interface – PreparedStatement - retrieving results - closing connection. Text / Reference T/R Book Title/Author Herbert Schildt, Dale Skrien: Java Fundamentals A Comprehensive T1 Introduction T2 Balaguruswamy E: Programming with Java, 6th edition. R1 Liang, Y Daniel: Introduction to JAVA Programming, Pearson, 9th Ed. R2 Herbert Schildt: Java The Complete Reference, Seventh Edition, R3 Herbert Schildt: Swing A Beginner's guide R4 Paul Deital, Harvey Deital: Java How to Program, R5 Yang HU: Easy Learning JDBC+Mysql Online Resources SI.No Website Link 1 https://onlinecourses.nptel.ac.in/noc20_cs08/course 2 https://www.tutorialspoint.com/java means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 4-□□ Develop simple application with database 4 Applying Database – Creation of databases-Tables -SQL commands like -create, alter, drop, select, insert, delete Database Connectivity – JDBC – RowSet interface – PreparedStatement - retrieving

results - closing connection. Text / Reference T/R Book Title/Author Herbert Schildt, Dale Skrien: Java Fundamentals A Comprehensive T1 Introduction T2 Balaguruswamy E: Programming with Java, 6th edition. R1 Liang, Y Daniel: Introduction to JAVA Programming, Pearson, 9th Ed. R2 Herbert Schildt: Java The Complete Reference, Seventh Edition, R3 Herbert Schildt: Swing A Beginner's guide R4 Paul Deital, Harvey Deital: Java How to Program, R5 Yang HU: Easy Learning JDBC+Mysql Online Resources Sl.No Website Link 1 https://onlinecourses.nptel.ac.in/noc20_cs08/course 2 <https://www.tutorialspoint.com/java> definition/meaning application. steps, application . Technical terms English- .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain 1 Understanding. Programming. Explain the JDK & IDE's for developing java. (3 marks)**

► **Define or explain 1 Understanding. applications Outline the general structure of a Java Understanding.. (3 marks)**

► **Define or explain program and explain steps to create, compile 1 and execute a program Describe objects and classes and use classes. (3 marks)**

► **Define or explain 2 Understanding. to model objects. Demonstrate how to define classes and create. (3 marks)**

Module 2

► **Define or explain 3 Understanding inheritance with examples. Illustrate invoking the superclass constructor. (3 marks)**

► **Define or explain 2 Applying and methods using super keywords. Illustrate overriding instance methods in the. (3 marks)**

► **Define or explain 2 Applying subclass.. (3 marks)**

- ▶ **Define or explain Illustrate polymorphism and dynamic binding. (3 marks)**

Module 3

- ▶ **Define or explain Outline GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI. (3 marks)**

- ▶ **Define or explain containers, components, event-handling classes 2 Applying. and interfaces to develop simple applications.. (3 marks)**

- ▶ **Define or explain Describe steps to create and manipulate GUI 3 Understanding. controls. Explain handling of mouse events and. (3 marks)**

- ▶ **Define or explain 2 Understanding. keyboard events. Explain Common GUI event type and Listener. (3 marks)**

Module 4

- ▶ **Define or explain Explain relational database & SQL. (3 marks)**

- ▶ **Define or explain Explain SQL commands.. (3 marks)**

- ▶ **Define or explain Illustrate the stages in JDBC 2 Understanding Illustrate the interfaces in JDBC -register the. (3 marks)**

- ▶ **Define or explain driver- connection- PreparedStatement- 2 Applying Executing statements- Resultset. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **1 Understanding**.

Programming. Explain the JDK & IDE's for developing java.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **3 Understanding inheritance**

with examples. Illustrate invoking the superclass constructor.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Outline GUI Programming in Java using swing. 2**

Understanding. Make use of packages containing GUI.

Module 4

1-mark definition: State the meaning of **Explain relational database & SQL.**

3-mark explanation: Explain one principle, process or comparison from

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- https://onlinecourses.nptel.ac.in/noc20_cs08/course
<https://www.tutorialspoint.com/java>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 5281. Verify institutional updates against the current official curriculum.